

INTRODUCTION TO BIRTH-DEATH MODELS

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Phylogeography Workshop

Day 2

1 BIRTH-DEATH MODEL

2 BIRTH-DEATH-SAMPLING MODEL

3 SAMPLING THROUGH TIME

OUTLINE

In these slides, we will learn about

- the birth-death model
- the birth-death sampling model

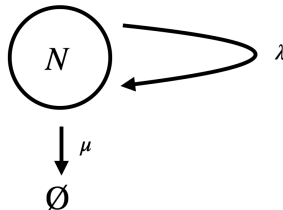
These both generalize the Yule model. The first one adds death; the second one also adds sampling through time

TABLE OF CONTENTS

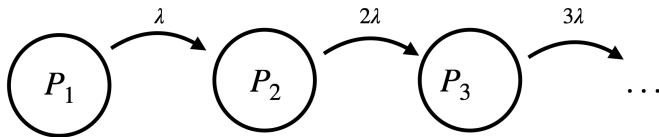
- 1 BIRTH-DEATH MODEL
- 2 BIRTH-DEATH-SAMPLING MODEL
- 3 SAMPLING THROUGH TIME

ADDING DEATH

What if individuals/lineages can also die at rate μ ?

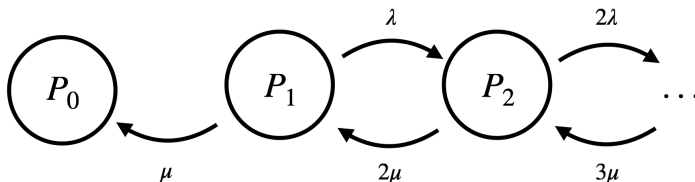


WITHOUT DEATH



In the Yule model, extinction is impossible

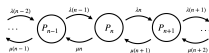
ADDING DEATH



In the birth-death model, extinction is possible: once we enter the state with 0 lineages, we are stuck – can't have birth from nothing

The state $n = 0$ is an **absorbing state** of this continuous-time Markov chain

FORWARD EQUATION FOR BIRTH-DEATH PROCESS



In this case, the Forward Equation will read

$$\frac{dp_{i,j}}{dt} = -p_{i,j}j(\lambda + \mu) + p_{i,j-1}(j-1)\lambda + p_{i,j+1}\mu(j+1) \quad \text{for } j \geq 1,$$

...

$$\frac{dp_{i,1}}{dt} = -p_{i,1}(\lambda + \mu) + p_{i,2}2\mu,$$

$$\frac{dp_{i,0}}{dt} = p_{i,1}\mu.$$

FORWARD EQUATION FOR BIRTH-DEATH PROCESS

Then given the Forward equation:

$$\frac{dp_{i,j}}{dt} = -p_{i,j}j(\lambda + \mu) + p_{i,j-1}(j-1)\lambda + p_{i,j+1}\mu(j+1) \quad \text{for } j \geq 1,$$

...

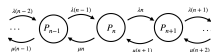
$$\frac{dp_{i,1}}{dt} = -p_{i,1}(\lambda + \mu) + p_{i,2}\mu,$$

$$\frac{dp_{i,0}}{dt} = p_{i,1}\mu.$$

we have an equation for $p_n = p_{1,n}$:

$$\frac{dp_n}{dt} = -p_n n(\lambda + \mu) + p_{n-1}(n-1)\lambda + p_{n+1}(n+1)\mu.$$

BACKWARD EQUATION FOR BIRTH-DEATH PROCESS



The Backward Equation will read

$$\frac{dp_{i,j}}{dt} = -i(\lambda + \mu)p_{i,j} + i\lambda p_{i+1,j} + i\mu p_{i-1,j} \quad \text{for } i \geq 1,$$

...

$$\frac{dp_{i,1}}{dt} = -(\lambda + \mu)p_{i,1} + i\lambda p_{i+1,1} + i\mu p_{i-1,1},$$

$$\frac{dp_{i,0}}{dt} = i\lambda p_{i+1,0} + i\mu p_{i-1,0}.$$

BACKWARD EQUATION FOR BIRTH-DEATH PROCESS

From this system:

$$\frac{dp_{i,j}}{dt} = -i(\lambda + \mu)p_{i,j} + i\lambda p_{i+1,j} + i\mu p_{i-1,j} \quad \text{for } i \geq 1,$$

...

$$\frac{dp_{i,1}}{dt} = -(\lambda + \mu)p_{i,1} + i\lambda p_{i+1,1} + i\mu p_{i-1,1},$$

$$\frac{dp_{i,0}}{dt} = i\lambda p_{i+1,0} + i\mu p_{i-1,0},$$

and considering $p_n = p_{1,n}$, we see that

$$\begin{aligned} \frac{dp_{1,n}}{dt} &= -(\lambda + \mu)p_{1,n} + \lambda p_{2,n} + 0, \\ &= -(\lambda + \mu)p_n + \lambda \sum_k p_k p_{n-k} \end{aligned}$$

because $p_{0,n} = 0$.

TREES IN THE BIRTH-DEATH PROCESS

Again, we can use the Forward/Backward equations to solve for $p_n(t)$, the probability there are n lineages at time t .

Exercise Solve the Birth-death model for $p_n(t)$. Hint: the generating function approach works, similar to our approach with the Yule model. The Method of Characteristics will still be useful to solve the PDE. You may want to look up “Riccati equation” to find a useful transformation to help you solve for the characteristic curves.

$p_n(t)$ IN THE BIRTH-DEATH PROCESS

Solution: In the phylodynamics literature, it is typical to reparameterize time with $t = 0$ corresponding to the present, and $t = T$ corresponding to the root. The branching times, x_i , are measured wrt this new parameterization.

If you work through the calculations, you should find that

$$p(x_1, x_2, \dots, x_{n-1} | T, \lambda, \mu) = (n-1)! \prod_{i=1}^{n-1} \frac{(\lambda - \mu)^2 e^{-(\lambda - \mu)x_i}}{(\lambda - \mu e^{-(\lambda - \mu)x_i})^2} \frac{\lambda - \mu e^{-(\lambda - \mu)T}}{1 - e^{-(\lambda - \mu)T}}$$

(The formula changes a little depending on whether one conditions on the total time T , or only on the time when the most recent common ancestor ($T_{\text{MRCA}} = x_1$) was alive.¹)

¹Tanja Stadler. “On incomplete sampling under birth–death models and connections to the sampling-based coalescent”. In: *Journal of theoretical biology* 261.1 (2009), pp. 58–66.

TABLE OF CONTENTS

1 BIRTH-DEATH MODEL

2 BIRTH-DEATH-SAMPLING MODEL

3 SAMPLING THROUGH TIME

THE PROBABILITY OF AN OBSERVED TREE²

The model: a fraction ρ of living lineages are sampled at the present time (but otherwise the tree is simulated from the birth-death model with birth rate λ , death rate μ)

Key difference with this formula: time is measured going from the tips to the root. Time $t = 0$ corresponds to the present, and time $t = T$ corresponds to root. The bifurcation times x_i are measured wrt this reparameterization:

$$p(x_1, x_2, \dots, x_{n-1}) = n!(\lambda - \mu)(\lambda\rho)^{n-1} \frac{e^{-(\lambda-\mu)x_1}}{\rho\lambda + (\lambda(1-\rho) - \mu)e^{-(\lambda-\mu)x_1}} \prod_{i=1}^{n-1} \frac{(\lambda - \mu)^2 e^{-(\lambda-\mu)x_i}}{(\rho\lambda + (\lambda(1-\rho) - \mu)e^{-(\lambda-\mu)x_i})^2}$$

²Tanja Stadler. “On incomplete sampling under birth–death models and connections to the sampling-based coalescent”. In: *Journal of theoretical biology* 261.1 (2009), pp. 58–66.

AN INVARIANT MANIFOLD IN PARAMETER SPACE

This model

$$p(x_1, x_2, \dots, x_{n-1}) = n!(\lambda - \mu)(\lambda\rho)^{n-1} \frac{e^{-(\lambda-\mu)x_1}}{\rho\lambda + (\lambda(1-\rho) - \mu)e^{-(\lambda-\mu)x_1}} \prod_{i=1}^{n-1} \frac{(\lambda - \mu)^2 e^{-(\lambda-\mu)x_i}}{(\rho\lambda + (\lambda(1-\rho) - \mu)e^{-(\lambda-\mu)x_i})^2},$$

has an **invariant manifold** in parameter space.

Consider a new sampling probability $\rho' \neq \rho$. Define the new parameters

$$\begin{aligned}\lambda' &= \rho\lambda/\rho', \\ \mu' &= \mu - \lambda(1 - \rho/\rho').\end{aligned}$$

With this reparameterization, $\lambda' - \mu' = \lambda - \mu$, $\rho'\lambda' = \rho\lambda$, and $\lambda'(1 - \rho') - \mu' = \lambda(1 - \rho) - \mu$.

This means that

$$p(\mathbf{x}|\lambda, \mu, \rho) = p(\mathbf{x}|\lambda', \mu', \rho').$$

THIS BIRTH-DEATH-SAMPLING MODEL IS STRUCTURALLY UNIDENTIFIABLE

This:

$$p(\mathbf{x}|\lambda, \mu, \rho) = p(\mathbf{x}|\lambda', \mu', \rho')$$

is a very nasty property. The fact that two different parameterizations ($(\lambda, \mu, \rho) \neq (\lambda', \mu', \rho')$) yield the same probability density for a tree means that we cannot uniquely estimate parameters using Maximum Likelihood.

This is an example of a **structurally unidentifiable model**.

It is **impossible** to estimate the parameters (λ, μ, ρ) together. The best we can do is fix one of the parameters and estimate the other two.

TABLE OF CONTENTS

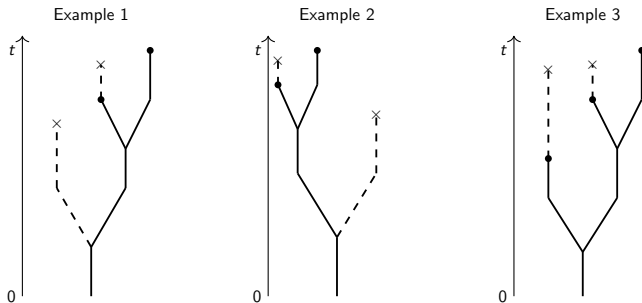
1 BIRTH-DEATH MODEL

2 BIRTH-DEATH-SAMPLING MODEL

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WHAT IF WE SAMPLE THROUGH TIME?

However, there is a possible remedy. In some applications, at least, we may collect samples at some rate through time, rather than just at the end:



— observed lineage --- unobserved lineage

● sampling × death

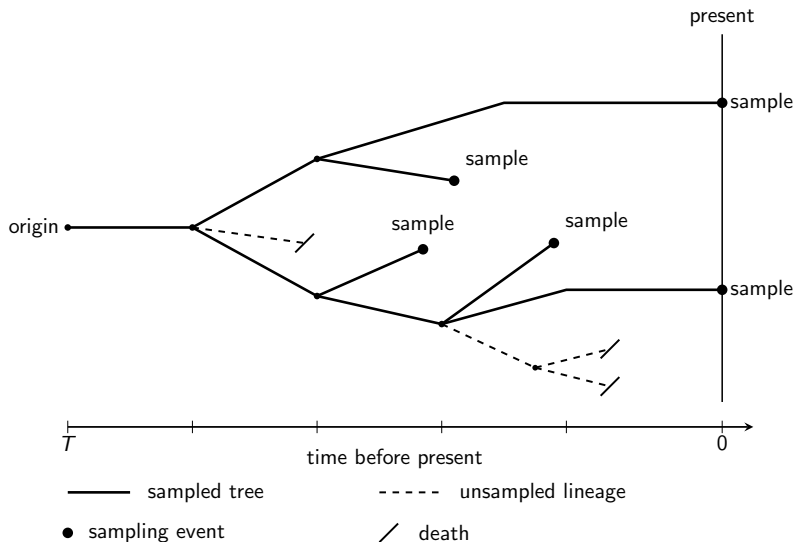
Will this change our ability to estimate parameters?

SAMPLING THROUGH TIME³

Let's consider an extension of the birth-death model (parameters (λ, μ)) with sampling occurring through time at rate ψ per-lineage

³Tanja Stadler. "Sampling-through-time in birth-death trees". In: *Journal of theoretical biology* 267.3 (2010), pp. 396–404.

SAMPLING-THROUGH-TIME BIRTH-DEATH MODEL



TREE LIKELIHOOD

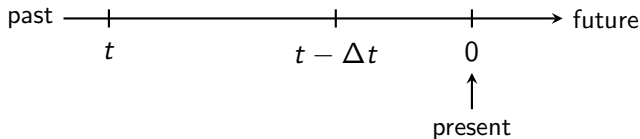
We now calculate tree likelihoods in a fundamentally different way from before.

Up to now, the quantity $p_n(t)$ has been tractable to obtain from the Kolmogorov Forward Equation for the transition probabilities.

We have seen that the probability of an individual tree's branching times marginalizes into $p_n(t)$

The **transition probabilities** $p_{ij}(t)$ were of the form $p_{1n}(t) = \mathcal{P}(n \text{ lineages at time } t \mid 1 \text{ lineage at time } 0)$.

This is intractable for heterochronous trees



We are going to switch things up and introduce a new timescale (which is more commonly used in the phylodynamics literature).

We set $t = 0$ to the present time, and have $t > 0$ indicate the amount of time into the past

The stochastic process generating trees is still defined from the past to the future. The most important rule of stochastic processes is: updates to the system at the present cannot use future information.

A NEW METHOD OF CALCULATION

We define two new quantities.

Time is measured going into the past now.

$$p_0(t) = \mathcal{P} \left(\begin{array}{c} \text{a lineage has no sampled} \\ \text{or extant descendants} \end{array} \middle| \text{the lineage is alive at time } t \right),$$

$$p_1(t) = \mathcal{P} \left(\begin{array}{c} \text{a lineage has exactly one} \\ \text{sampled extant descendant and} \\ \text{no sampled extinct descendant} \end{array} \middle| \text{the lineage is alive at time } t \right).$$

These are not like the $p_n(t)$'s we have seen up to now. Those were transition probabilities (really of the form $p_{1n}(t)$).

These new objects are **not** transition probabilities on the number of lineages. In the parlance of Mathematical Finance, these objects might be called “expected future rewards”.

A NEW METHOD OF CALCULATION

$$p_0(t) = \mathcal{P} \left(\begin{array}{c} \text{a lineage has no sampled} \\ \text{or extant descendants} \end{array} \middle| \text{the lineage is alive at time } t \right),$$
$$p_1(t) = \mathcal{P} \left(\begin{array}{c} \text{a lineage has exactly one} \\ \text{sampled extant descendant and} \\ \text{no sampled extinct descendant} \end{array} \middle| \text{the lineage is alive at time } t \right).$$

Notice where the variable t shows up. It's in the conditional part of these expressions.

Contrast this with our transition probabilities we had before (and using our new convention that time increments going backward into the past):

$$p_{1n}(t) = \mathcal{P}(n \text{ lineages at time } t \mid 1 \text{ lineage at time } T).$$

A NEW METHOD OF CALCULATION

Forward Equations tend to be useful for this type of object:

$$\mathcal{P}(\dots t | \dots),$$

Backward Equations tend to be useful for this type of object:

$$\mathcal{P}(\dots | \dots t).$$

A NEW METHOD OF CALCULATION

Consequently, we derive **Backward Equations** for the quantities

$$p_0(t) = \mathcal{P} \left(\begin{array}{c} \text{a lineage has no sampled} \\ \text{or extant descendants} \end{array} \middle| \text{the lineage is alive at time } t \right),$$

$$p_1(t) = \mathcal{P} \left(\begin{array}{c} \text{a lineage has exactly one} \\ \text{sampled extant descendant and} \\ \text{no sampled extinct descendant} \end{array} \middle| \text{the lineage is alive at time } t \right).$$

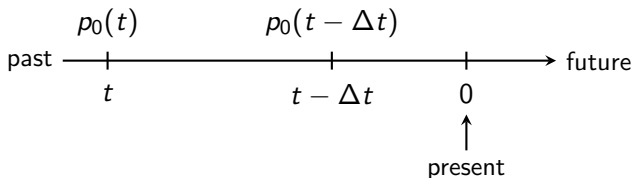
past $p_0(t)$ $p_0(t - \Delta t)$ future

t $t - \Delta t$ 0

present

It is easy to get confused with the derivations to follow. The stochastic process updates from past $\rightarrow$ present, but our time coordinate increments from the present to the past

A NEW METHOD OF CALCULATION



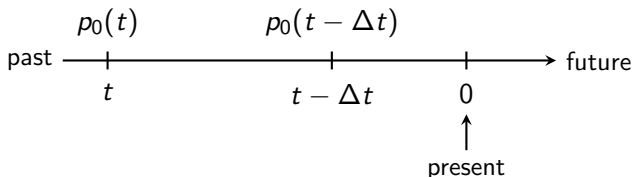
We use the total law of probability to relate $p_0(t)$ to $p_0(t - \Delta t)$:

$$\underbrace{p_0(t)}_{P(\text{dead by end}|\text{alive at } t)} = \underbrace{\mu\Delta t}_{P(\text{death})} \cdot \underbrace{1}_{P(\text{dead by end}|\text{just died})} + \underbrace{\lambda\Delta t}_{P(\text{birth})} \cdot \left(\underbrace{p_0(t - \Delta t)}_{P(\text{dead by end}|\text{still alive})} \right)^2 + \underbrace{(1 - (\mu + \lambda + \psi)\Delta t)}_{P(\text{nothing happens})} \cdot \underbrace{p_0(t - \Delta t)}_{P(\text{dead by end}|\text{still alive})}$$

The Law of Total Probability says

$$\mathcal{P}(A) = \mathcal{P}(B)\mathcal{P}(A|B) + \mathcal{P}(C)\mathcal{P}(A|C)$$

A NEW METHOD OF CALCULATION



We arrange this into an equation involving the difference quotient of $p_0(t)$:

$$\begin{aligned}
 p_0(t) &= \mu \Delta t \times 1 + \lambda \Delta t p_0(t - \Delta t)^2 + (1 - (\mu + \lambda + \psi) \Delta t) p_0(t - \Delta t), \\
 \implies \\
 p_0(t) - p_0(t - \Delta t) &= (\mu + \lambda p_0(t - \Delta t)^2 - (\mu + \lambda + \psi) p_0(t - \Delta t)) \Delta t, \\
 \implies \\
 \frac{p_0(t) - p_0(t - \Delta t)}{\Delta t} &= (\mu + \lambda p_0(t - \Delta t)^2 - (\mu + \lambda + \psi) p_0(t - \Delta t)).
 \end{aligned}$$

A NEW METHOD OF CALCULATION

Then, in the limit $\Delta t \rightarrow 0$,

$$\frac{p_0(t) - p_0(t - \Delta t)}{\Delta t} \rightarrow \frac{dp_0}{dt},$$

and likewise

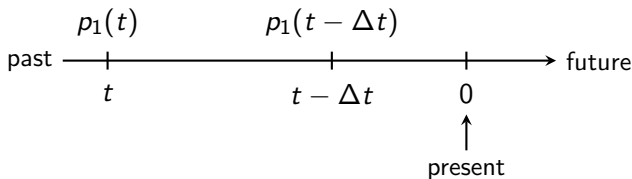
$$\begin{aligned} & \mu + \lambda p_0(t - \Delta t)^2 - (\mu + \lambda + \psi)p_0(t - \Delta t) \\ & \rightarrow \\ & \mu + \lambda p_0(t)^2 - (\mu + \lambda + \psi)p_0(t). \end{aligned}$$

So

$$\frac{dp_0}{dt} = \mu + \lambda p_0(t)^2 - (\mu + \lambda + \psi)p_0(t).$$

Exercise Try repeating this derivation with the time variable increasing forward into the present instead. Should the ODE to change depending on which time convention you use?

Now, DO $p_1(t)$:



We use the total law of probability to relate $p_1(t)$ to $p_1(t - \Delta t)$:

$$\underbrace{p_1(t)}_{P(\text{one descendant by end} \mid \text{alive at } t)} = \underbrace{2 \lambda \Delta t}_{P(\text{birth})} \cdot \underbrace{p_0(t - \Delta t)}_{P(\text{dead by end} \mid \text{still alive})} \cdot \underbrace{p_1(t - \Delta t)}_{P(\text{one descendant by end} \mid \text{still alive})} + \underbrace{(1 - (\mu + \lambda + \psi)\Delta t)}_{P(\text{nothing happens})} \cdot \underbrace{p_1(t - \Delta t)}_{P(\text{one descendant by end} \mid \text{still alive})}$$

DIFFERENCE QUOTIENT FOR p_1

The equation for the difference quotient is

$$\frac{p_1(t) - p_1(t - \Delta t)}{\Delta t} = 2\lambda p_0(t - \Delta t)p_1(t - \Delta t) - (\lambda + \mu + \psi)p_1(t - \Delta t),$$

and the limit $\Delta t \rightarrow 0$ we get

$$\frac{dp_1}{dt} = 2\lambda p_0 p_1 - (\lambda + \mu + \psi)p_1.$$

BACKWARD EQUATIONS FOR p_0 AND p_1

We now have backward equations for the probability of zero lineages by the present time, and the probability of exactly one descendant by the present time, given a lineage is alive at time t in the past:

$$\begin{aligned}\frac{dp_0}{dt} &= \mu + \lambda p_0^2 - (\mu + \lambda + \psi)p_0, \\ \frac{dp_1}{dt} &= 2\lambda p_0 p_1 - (\lambda + \mu + \psi)p_1.\end{aligned}$$

In addition to sampling through time, lineages at the end (time $t = 0$) may be sampled with probability ρ . Hence, we supplement these ODEs with the initial conditions

$$\begin{aligned}p_0(0) &= 1 - \rho, \\ p_1(0) &= \rho.\end{aligned}$$

BACKWARD EQUATIONS FOR p_0 AND p_1

$$\frac{dp_0}{dt} = \mu + \lambda p_0^2 - (\mu + \lambda + \psi)p_0,$$

$$\frac{dp_1}{dt} = 2\lambda p_0 p_1 - (\lambda + \mu + \psi)p_1,$$

$$p_0(0) = 1 - \rho,$$

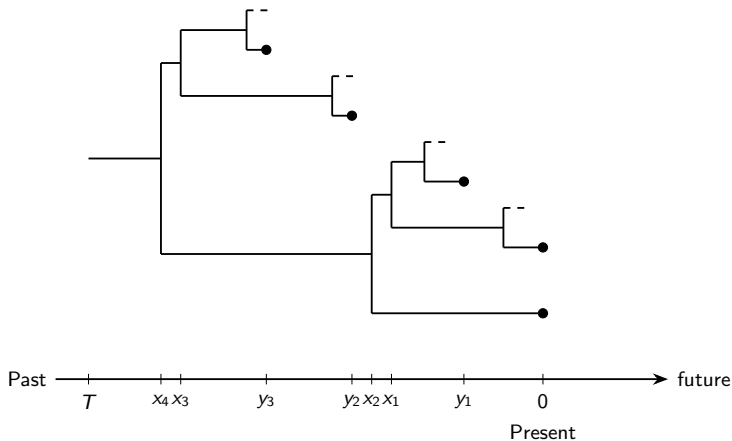
$$p_1(0) = \rho.$$

There is an analytical solution to this system⁴, but we won't work through that here.

Assuming we have $p_0(t)$ and $p_1(t)$, we can build up the probability of an entire tree using a recursive algorithm:

⁴Tanja Stadler. "Sampling-through-time in birth-death trees". In: *Journal of theoretical biology* 267.3 (2010), pp. 396–404.

FULL TREE AND SAMPLED LINEAGES



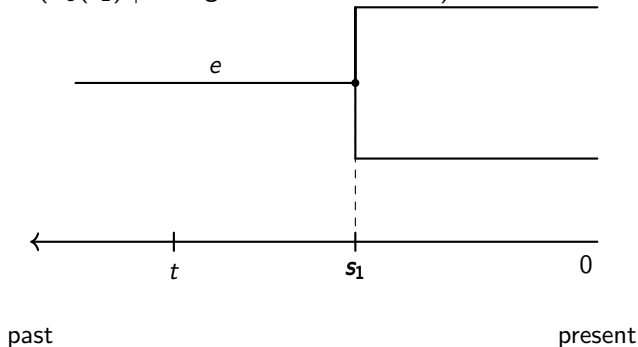
Branching times in the observed tree occur at times x_k

Sampling events occur at times y_k (at rate ψ) or at time 0 (with probability ρ)

PROBABILITY OF A SUBTREE

$\mathcal{T}_e(s_1)$: the sub-tree descending from edge e at time s_1

$g_e(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive at time } t)$

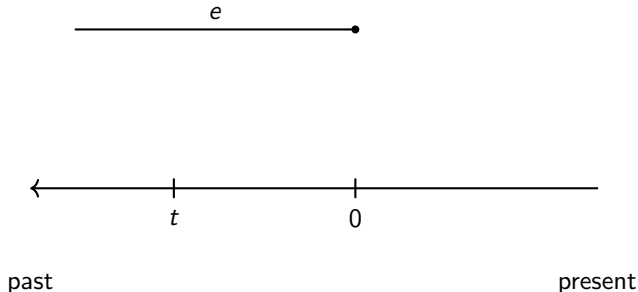


$g_e(t)$ satisfies the backward equation

$$\frac{dg_e}{dt} = -(\lambda + \mu + \psi)g_e(t) + 2\lambda p_0(t)g_e(t).$$

FOUR TYPES OF INITIAL CONDITIONS - FIRST TYPE:

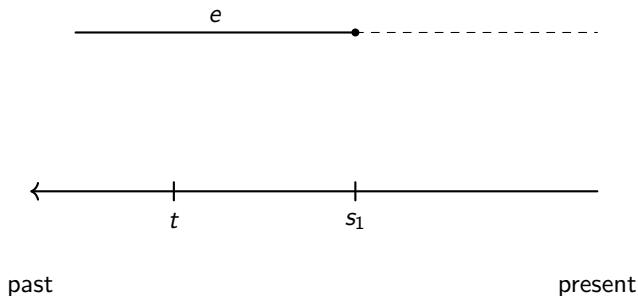
Case 1: edge e is a terminal edge with sampled tip at the present ($t = 0$):



$g_e(0) = \rho$; integrating dg_e/dt gives $p_1(t)$ up until the next node

SECOND TYPES OF INITIAL CONDITION

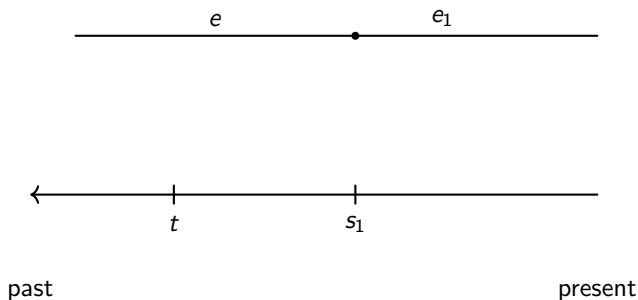
Case 2: edge e is sampled before the present ($s_1 > 0$), and its descendant edge produces no sampled descendants



$$g_e(s_1) = \psi p_0(s_1)$$

THIRD TYPE OF INITIAL CONDITION

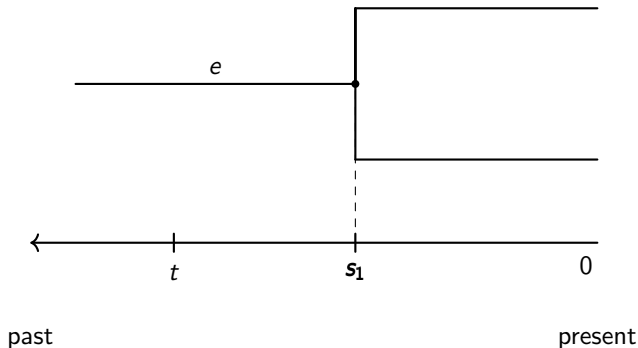
Case 3: edge e is sampled before the present ($s_1 > 0$), and its descendant edge **does** produce sampled descendants



$$g_e(s_1) = \psi g_{e_1}(s_1)$$

FOURTH TYPE OF INITIAL CONDITION

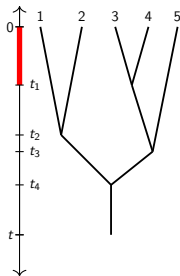
Case 4: a lineage e branches into two lineages e_1 and e_2 , both of which produce sampled descendants:



$g_e(s_1) = \lambda g_{e_1}(s_1) g_{e_2}(s_1)$ (recall that new lineages emerge at rate λ per lineage)

BACK TO THE YULE MODEL:

Exercise Apply this algorithm for an example ultrametric Yule tree ($\mu = 0$, $\psi = 0$, $\rho = 1$, $\lambda > 0$), and determine whether you obtain the same tree probability that we obtained in the previous lecture.



In this case, $p_0 = 0$ (why?), and p_1 satisfies the backward equation

$$\frac{dp_1}{dt} = -\lambda p_1,$$

using the convention that the time variable, t , increases into the past.

What initial conditions should we use?

YOUR MISSION

In the accompanying demo, you will simulate birth-death and birth-death sampling trees, and try to estimate parameters by fitting the models to trees using Maximum Likelihood.

In the specific case of birth-death-sampling trees, we will explore whether our ability to fit models depends on whether lineages are all sampled at the same time ($t = 0$), or over an interval of time.